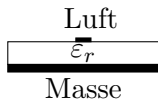


HF

1 Grundlagen

Mikrostreifenleitungen: Quasi-TEM-Wellen



Triplate Leitung: TEM-Wellen



Leitungsgleichung γ, Z_L

$$U = U_f e^{-\gamma z} + U_b e^{\gamma z}$$

$$I = \frac{U_f}{Z_L} e^{-\gamma z} - \frac{U_b}{Z_L} e^{\gamma z}$$

Normierungs:

$$u = \frac{U}{\sqrt{Z_L}}, i = I \sqrt{Z_0}, U_f = I_f Z_L$$

$$u = u_f e^{-\gamma z} + u_b e^{\gamma z} = a + b$$

$$i = a - b$$

Matrixbeschreibungen

- S-Parameter (Streuparameter):**

- Beschreibung über Wellen (a_i (Eingang Oben), b_i (Ausgang unter), Nummer sind die Seite)

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

- * $S_{11}|_{a_2=0} = \frac{b_1}{a_1}$: Reflexion Eingang

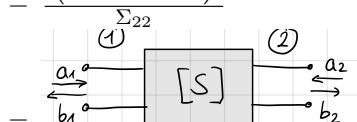
- * $S_{21}|_{a_2=0} = \frac{b_2}{a_1}$: Transmission (Verstärkung)

- * S_{12} : Rückwirkung

- * S_{22} : Reflexion Ausgang

$$\begin{pmatrix} A+B-C-D & 2\Delta_A \\ 2 & -A+B-C+D \end{pmatrix}$$

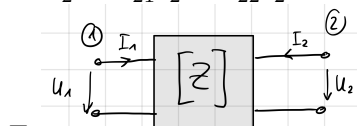
$$\begin{pmatrix} \Sigma_{12} & \Delta_{\Sigma} \\ 1 & -\Sigma_{21} \end{pmatrix}$$



- Impedanzparameter:**

- $U_1 = Z_{11} I_1 + Z_{12} I_2$

- $U_2 = Z_{21} I_1 + Z_{22} I_2$

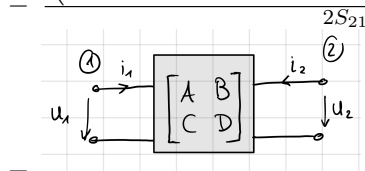


- Kettenparameter:**

- Darstellung der ABCD-Parameter

$$\begin{pmatrix} U_1 \\ I_1 \end{pmatrix} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} U_2 \\ -I_2 \end{pmatrix}$$

$$\begin{pmatrix} -\Delta_S + S_{11} - S_{22} + 1 & \Delta_S + S_{11} + S_{22} + 1 \\ \Delta_S - S_{11} - S_{22} + 1 & -\Delta_S - S_{11} + S_{22} + 1 \end{pmatrix}$$

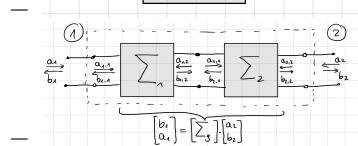
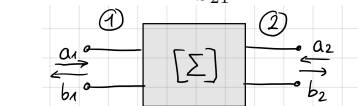


- Transmissionsmatrix:**

- Multiplikation Vereinfachung

$$\begin{pmatrix} b_1 \\ a_1 \end{pmatrix} = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} a_2 \\ b_2 \end{pmatrix}$$

$$1 \begin{pmatrix} -\Delta_S & S_{11} \\ -S_{22} & 1 \end{pmatrix}$$



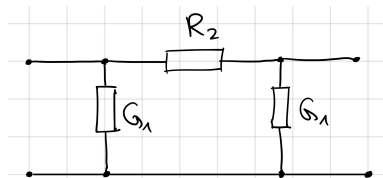
$$[\Sigma_g] = [\Sigma_1][\Sigma_2]$$

1.1 Leitungen

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & \frac{R}{Z_0} \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ GZ_0 & 1 \end{pmatrix}$$

1.1.1 Dämpfungsglied



$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ G_1 Z_0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \frac{R_2}{Z_0} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ G_1 Z_0 & 1 \end{pmatrix}$$

2 Koppler

Taktanregung

- Gleichtaktanregung:** $\oplus$ Signale gleicher Amplitude und gleicher Phase.

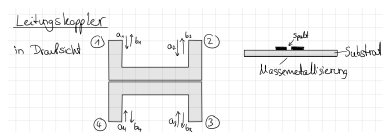
- Z_e
- $a_1^+ = \frac{1}{2} a_1, a_4^+ = \frac{1}{2} a_1$

- Gegentaktanregung:** $\ominus$ Signale gleicher Amplitude und entgegengesetzter Phase.

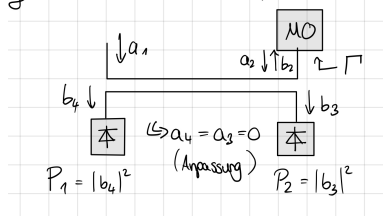
- Z_d
- $a_1^- = \frac{1}{2} a_1, a_4^- = -\frac{1}{2} a_1$

- Überlagerung: $a_1^+ + a_1^- = a_1, a_4^+ + a_4^- = 0$

2.1 Leitungskoppler

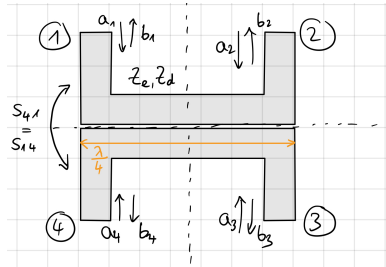


$$g_{es}: \Gamma = f([S], P_1, P_2)$$



$$|\Gamma|^2 = \frac{P_2}{P_1} \left| \frac{S_{41}}{S_{32} S_{21}} \right|^2$$

Zweifach Symmetrischer Zweitor:



Allseit. Anpassung:

$S_{11} = S_{22} = S_{33} = S_{44} = 0$

Symmetrien und Reziprok:

$S_{41} = S_{14} = S_{32} = S_{23}$

$S_{12} = S_{21} = S_{34} = S_{43}$

Mit Gleich- und Gegentaktanregung:

$b_1 = b_1^+ + b_1^- = S_{11}^+ a_1^+ + S_{11}^- a_1^- = \frac{1}{2}(S_{11}^+ + S_{11}^-) a_1, \rightarrow S_{11}$

$b_2 = b_2^+ + b_2^- = S_{21}^+ a_1^+ + S_{21}^- a_1^- = \frac{1}{2}(S_{21}^+ + S_{21}^-) a_1, \rightarrow S_{21}$

$b_3 = b_2^+ + b_2^- = S_{21}^+ a_4^+ + S_{21}^- a_4^- = \frac{1}{2}(S_{21}^+ - S_{21}^-) a_1, \rightarrow S_{31}$

$b_4 = b_4^+ + b_4^- = S_{11}^+ a_4^+ + S_{11}^- a_4^- = \frac{1}{2}(S_{11}^+ - S_{11}^-) a_1, \rightarrow S_{41}$

Normierung der Widerstände: $z_e = \frac{Z_e}{Z_0}, z_d = \frac{Z_d}{Z_0}$

$$\begin{bmatrix} A^+ & B^+ \\ C^+ & D^+ \end{bmatrix} = \begin{bmatrix} \cos(\beta l) & j z_e \sin(\beta l) \\ j \frac{1}{z_d} \sin(\beta l) & \cos(\beta l) \end{bmatrix}$$

$S_{11}^+ = \frac{A^+ + B^+ - C^+ - D^+}{A^+ + B^+ + C^+ + D^+} = \frac{j z_e \sin(\beta l) - j \frac{1}{z_d} \sin(\beta l)}{2 \cos(\beta l) + j(z_e + \frac{1}{z_d}) \sin(\beta l)}$

$S_{11}^- = \frac{j(z_d - \frac{1}{z_e}) \sin(\beta l)}{2 \cos(\beta l) + j(z_d + \frac{1}{z_e}) \sin(\beta l)}$

$S_{21}^+ = \frac{2}{A^+ + B^+ + C^+ + D^+} = \frac{2}{2 \cos(\beta l) + j(z_e + \frac{1}{z_d}) \sin(\beta l)}$

$S_{21}^- = \frac{2}{A^- + B^- + C^- + D^-} = \frac{2}{2 \cos(\beta l) + j(z_d + \frac{1}{z_e}) \sin(\beta l)}$

Allseitige anpassung

$S_{11} = 0 \rightarrow S_{11}^+ = -S_{11}^- \rightarrow z_e = \frac{1}{z_d} \leftrightarrow z_e z_d = 1 \rightarrow Z_e Z_d = Z_0^2$

Entkoppelte Tore: $S_{31} = \frac{1}{2}(S_{21}^+ - S_{21}^-) = 0 \rightarrow S_{21}^+ = S_{21}^-$

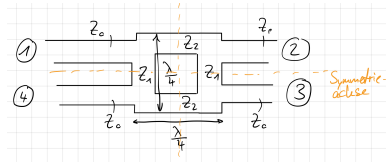
$S_{21} = S_{21}^+ = \frac{2}{2 \cos(\beta l) + j(z_e + \frac{1}{z_e}) \sin(\beta l)}$

$S_{41} = \frac{1}{2}(S_{11}^+ - S_{11}^-) = S_{11}^+ = \frac{j(z_e - \frac{1}{z_e}) \sin(\beta l)}{2 \cos(\beta l) + j(z_e + \frac{1}{z_e}) \sin(\beta l)}$

$\angle(S_{21}, S_{41}) = 90^\circ$

Mehrstufig wenn höhere Breitband.

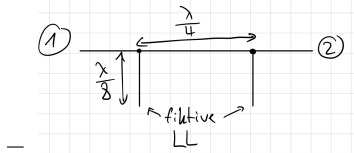
2.2 Ringkoppler



- 3dB-Kopplung: 1 → 2 und 1 → 3
- Entkopplung: 1 und 4
- Einspeisung an 1
- $Z_1 = ?, Z_2 = ?$

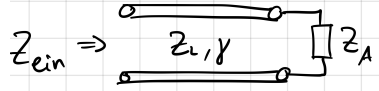
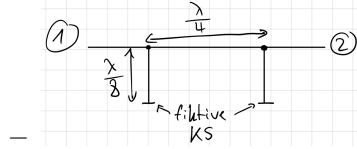
• Gleichtakt

– $\oplus$ an 1 und 4: $a_1^+ = \frac{1}{2} a_1, a_4^+ = \frac{1}{2} a_1$



• Gegentakt

– $\ominus$ an 1 und 4: $a_1^- = \frac{1}{2} a_1, a_4^- = -\frac{1}{2} a_1$



- $Z_{ein} = Z_L \frac{Z_A + j Z_L \tan(\beta l)}{Z_L + j Z_A \tan(\beta l)}$
- $\tan(\beta l) = \tan(\frac{2\pi}{\lambda} \frac{\lambda}{8}) = \tan(\frac{\pi}{4}) = 1$
- $\oplus Z_A \rightarrow \infty \Rightarrow Z_{ein,LL}^+ = \lim_{Z_A \rightarrow \infty} Z_{ein} = -j Z_L = -j Z_1$
- $\ominus Z_A \rightarrow 0 \Rightarrow Z_{ein,LL}^- = j Z_L = j Z_1 = j \frac{1}{Y_1}$

Kettenmatrizen fortsetzung

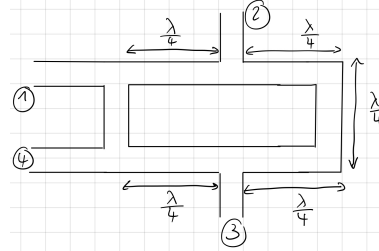
$$[M^+] = \begin{bmatrix} 1 & 0 \\ -j Y_1 Z_0 & 1 \end{bmatrix} \begin{bmatrix} 0 & j \frac{Z_2}{Z_0} \\ j \frac{Z_0}{Z_2} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -j Y_1 Z_0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} +Z_2 Y_1 & j \frac{Z_2}{Z_0} \\ j \frac{Z_0}{Z_2} - j Y_1^2 Z_0 Z_2 & +Y_1 Z_2 \end{bmatrix} = \begin{bmatrix} A^+ & B^+ \\ C^+ & D^+ \end{bmatrix}$$

Anforderung: Allseitige Anpassung

- $S_{11} = \frac{1}{2}(S_{11}^+ + S_{11}^-) \stackrel{!}{=} 0$
- $S_{41} = S_{14} = \frac{1}{2}(S_{11}^+ - S_{11}^-) \stackrel{!}{=} 0$
- $S_{11}^+ = \frac{A^+ + B^+ - C^+ - D^+}{A^+ + B^+ + C^+ + D^+} \stackrel{!}{=} 0$
- Da $A^+ = D^+ \Rightarrow B^+ = C^+$
- $\frac{Z_2}{Z_0} = \frac{Z_0}{Z_2} - Y_1^2 Z_2 Z_0$
- $|S_{21}| = |\frac{1}{2}(S_{21}^+ + S_{21}^-)| = |S_{31}| = |\frac{1}{2}(S_{21}^+ - S_{21}^-)|$
- $\frac{Z_2}{Z_0} = Y_1 Z_2 \Rightarrow Y_1 = \frac{1}{Z_0}$
- $1 = \frac{Z_0^2}{Z_2^2} - Y_1^2 Z_0^2 \Rightarrow Z_2 = \frac{1}{\sqrt{2}} Z_0$
- $\angle(S_{21}, S_{31}) = 90^\circ$
- 3dB-Kopplung: 1 → 2 und 1 → 3
- Entkopplung: 1 und 4

2.2.1 Rat-Race-Koppler



- Einsp. 1: 3 entkoppelt, $\angle(S_{21}, S_{41}) = 0^\circ$
- Einsp. 3: 1 entkoppelt, $\angle(S_{23}, S_{43}) = 180^\circ$